Eight Efficient Open Source Tools Sysadmins Can Use to Build VPNs

A virtual private network (VPN) acts as a secure tunnel, which extends over a public or shared network, so that data can be exchanged anonymously and securely across the Internet, just like using a private network. In this article, sysadmins can get an insight into the best free and open source VPN tools available.

With the explosive growth of the Internet, the one main thrust of every IT organisation is to exploit it to the maximum to further business interests. Initially, companies used the Internet to promote their products, brand image and services through corporate websites. But the Internet today is visualised as a platform with limitless possibilities. The focus has shifted to e-business, whereby the global reach of the Internet is used for easy access to key business applications and data that resides in traditional IT systems. Companies are looking for the best solution to securely and cost-effectively extend the reach of their applications and data across the world. While Web-enabled applications can be used to achieve this, a virtual private network (VPN) offers a more comprehensive and secure solution.

VPNs securely convey information across the Internet connecting remote users, branch offices and business partners to an extended corporate network. The basics of a VPN are highlighted in Figure 1.

According to a report by Orbis Research, the global VPN market was valued at US$ 15.64 billion in 2016 and is expected to reach US$ 35.73 billion by the end of 2022, growing at a CAGR of 14.77 per cent between 2016 and 2022.

A VPN is an extension of an enterprise's private intranet across a public network such as the Internet, creating a secure private connection, essentially through a private tunnel.

The VPN definition can be split as follows.

- *It is virtual:* This means that the physical infrastructure of the network has to be transparent to any VPN connection. In most cases, it also means that the physical network is not owned by the user of a VPN, but is a public network shared with many other users. To facilitate this transparency to the upper layers, protocol tunnelling techniques are used. To overcome the implications of not owning the physical network, service level agreements (SLAs) with network providers are created to provide, in the best possible manner, the performance and availability requirements needed by the VPN.
- *It is private:* The word 'private' with regard to VPNs refers to the privacy of the traffic that flows over the VPN network. VPN traffic flows over public networks and therefore the necessary security precautions need to be taken. The security requirements are: data encryption, data authentication, secure cryptographic keys to follow encryption and authentication, defence against the relay of packets and address spoofing.
- *It is a network:* A VPN is regarded as an extension to the company's existing network infrastructure. This means that it must be available to the rest of the network, to all or a specified subset of its devices and applications, by regular means of topology like routing and addressing.

Types of VPNs

VPNs can be classified into the following two main categories:

- Site-to-site VPNs
- Remote access VPNs

Site-to-site VPN: This allows